

other aspects of the curriculum.”

All the student teachers claimed that they would definitely consider using the outdoor environment to support their teaching of mathematics in the future. One student teacher enthusiastically claimed,

“It has made me realise how maths can be used in a fun way and in a meaningful way. I will definitely try to make one up for outdoors when I am teaching!”

Even though the activities were based at the Folk Park, student teachers had previously been exposed to a wide range of ideas and they felt they had gained a greater confidence and competence to plan, and prepare, mathematical activities for children at other outdoor locations. One student teacher commented,

“The visit has given me ideas to implement maths in different ways within the school context.”

Some student teachers did express concerns over taking a class of children outdoors in the future. The main area of concern appeared to centre on the management of children’s behaviour. However, being aware of opportunities for outdoor learning in mathematics at various locations such as the Ulster American Folk Park provided some reassurance, and students commented that they would certainly consider promoting the use of such centres when they are teaching mathematics in schools.

Overall, the off-campus visit allowed student teachers to gain first-hand experience of how the outdoor environment can be used to support and enhance children’s mathematical development. In addition, they learned how real-life historical contexts can be used to connect learning across the curriculum. The visit offered student teachers the opportunity to link theory with practice and trial a novel teaching approach in a supportive environment.

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# THE CONSTRUCTIVIST MATHEMATICS CLASSROOM

Jones et al apply the constructivist principles to a classroom incident.

By examining how people learn, the educational theories of Dewey, Piaget, Vygotsky and Bruner can be synthesized to give this set of core Constructivist principles.

Principles of effective mathematics teaching:

- 1 allows learning that is *active* and *reflective*. Students are required to transfer key concepts to new situations.
- 2 allows learning that is continuous. It is designed to allow students to understand new experiences in the light of *prior knowledge*.
- 3 allows learning that is social. It requires *interaction* to develop a deep conceptual understanding, positive relationships and a classroom community.

Here we examine a case study of Ms. Parker, a 7th grade middle school mathematics teacher. Let us consider her introductory lesson on surface area, in the light of these Constructivist beliefs.

9:00am – students enter the classroom and pick up their investigation. Ms Parker greets them, “Good morning class!” – “Welcome.”

Ms Parker gives a review of the day’s learning target: “*I can describe the perimeter and area of a cubic flat pattern.*” Students have started a new unit on 3-D geometry and are working towards investigations on surface area, and volume. Ms. Parker tells students that by the end of today’s lesson she will have evidence that each of them can calculate the area and perimeter of a cubic flat pattern, and have thought deeply about the properties of a cube. She explains that students will build on these understandings in future lessons as they consider relationships between the calculations of 2-D flat patterns, and the 3-D properties of cubes. Students

listen and several ask questions. Ms. Parker asks students to look at a set of objects.

Ms. Parker says "many things on this table look familiar." Students comment to each other as they recognize moving boxes, gift boxes, tissue boxes, and other everyday items. Ms. Parker asks: "Does anyone notice anything these items have in common?"

"They are all small?" offers Kate

"That's true. Good start," Ms. Parker comments. "What else?"

Nisha suggests "They are all cube shaped?"

"Terrific. That's what we are looking for.

Manufacturers chose a cubic box for their product. This could just be coincidence, but as problem solvers we need to ask, why? So, discuss with your team, and make a list of reasons why cubic boxes might be chosen."

Students work collectively, documenting their ideas while developing a purpose for their study. Most ideas are centered around four hypotheses.

Manufacturers are looking to:

- 1 minimize the material needed to make each box
- 2 maximize the space for product
- 3 create an ideal shape for transport, storage and display
- 4 design a box with aesthetic appeal to customers.

Ms. Parker explains how these hypotheses will form the basis of future work.

"Now we are going to consider the first idea we heard – how do we make a box that holds what we need, that doesn't use lots of cardboard? Package designers have options, we will now consider those options and decide which boxes are best and why."

"Before we get started we need to know how boxes are made. Ms. Parker projects a picture of a manufacturing plant. "Boxes start as sheets of cardboard that are cut, folded into a box shape, then glued or taped together. You will do that today, on a smaller scale." Students turn their investigations over to find 10 cubic nets and the instruction:

"How many of the flat patterns will fold into a box shaped like a unit cube? Sketch each pattern you find, then test each pattern by cutting and folding it into a box."

Students are given twelve minutes to construct the nets with their team and decide which make, cubes and which don't. With nets that make cubes, students find the perimeter and area, looking for patterns and relationships. They work in groups, share materials, exchange ideas and question each other.

Ms. Parker overhears:

Terrell "Does there always have to be six boxes in the net?"

Maurice "I think so. But I am not sure why"

After ten minutes with the whole class, Ms.

Parker discusses the 'hard parts' and ensures students have documented their ideas. With five minutes remaining, students complete 3 of 5 reflection prompts which they submit to Ms. Parker as they leave. These prompts provide formative assessment data for Ms. Parker and require students to consider both the products they produced, and the thinking and interaction that took place. As students leave, she thanks them for their contributions to their group and wishes them a nice rest of the day.

Now we consider the instructional choices made by Ms. Parker and how they align with the learning principles established. It is evident that student learning in this class is *active*, *reflective*, *continuous* and *social*. Our aim is to pinpoint Ms. Parker's role.

### Principle 1

In *Experience & Education*, Dewey states that quality education arises from experiences, which are both *active* and *reflective*. *Active* experiences require students to use what they know to discover, challenge, and ponder new ideas. In active learning, students create, manipulate and generate new ideas by putting them to use in a way which is novel and meaningful. Deep understanding takes place when a series of connected and meaningful learning opportunities are aligned to previous experience, and learners have frequent opportunities to reflect on both prior and current understandings. By the mere design and delivery of the material, good instruction stimulates curiosity, creates purpose, and promotes further learning (Dewey, 1937).

Investigating surface area in the context of packages that maximize the volume, and minimize the surface area, fulfills both parts of Dewey's definition of quality education. With regard to Dewey's assertion that quality education is active, Ms. Parker's lesson relied on student participation and discussion. Allowing students to work in whole groups, small groups and individually, made the learning valuable and worthwhile. At no point were students told to follow a *procedure*, or *perform mindless calculations*. Students constructed their own meaning of surface area by manipulating ideas, recording observations, and drawing conclusions. The discussion between Maurice and Terrell, shows that students were not only *active* in answering questions, but also active in generating questions for further study. This indicates that rather than memorizing information students were generating their own meaning.

Ms. Parker's methods also fulfill the *reflection*

aspect of Dewey's ideal learning experience since this lesson had a real life context. The lesson began and ended with a series of student reflections, at the beginning to shape the lesson, and at the end provided closure, and directions for further inquiry. *Reflection* provided formative assessment, self assessment for the learners, motivation, and interest.

## Principle 2

In *Towards a Theory of Instruction* (1966), Bruner states that effective instruction is engaging and stimulating to student minds. In inquiry-based learning, students use what they already know, and have experienced; in order to construct knowledge in a way that is personally meaningful. He states that effective and highly engaging teachers structure their lessons in a way that students can build upon their *prior knowledge* and revisit previous understandings in a spiral organisation. Contrary to most instruction, effective instruction is not a series of fragmented 40-minute lessons, but a *continuous* and gradual change in schema (Piaget, 1972) fostered by considering the interconnectedness of learning experiences.

Ms Parker exemplifies this second principle by deliberately relating each new concept, skill and procedure to one previously mastered. Students' understanding of perimeter and area are spiraled into the lesson so they can easily relate surface area with prior understanding of perimeter. Ms. Parker assumes that students already have an intuitive schema of surface area, but also realizes they probably lack the sophisticated mathematical vocabulary needed to talk about it. Through the manipulation of nets, and contemplation of the guiding question "Why are so many boxes cubic?" Ms. Parker invites students to assimilate the new concept of surface area into their existing schema for perimeter, thereby building upon *prior knowledge*, instead of trying to fabricate original thought.

Ms. Parker intentionally foreshadows where students are heading in subsequent lessons, and how they will use this new information. By setting a purpose for learning, Ms. Parker not only identifies students' conceptual change as evidence of their work in today's class, but also uses what they have learned to motivate further exploration (Bruner, 1966).

## Principle 3

According to Vygotsky's theory of socio-cultural development (1977), students learn best when learning experiences involve *social interaction* with peers and adults. Since *social interaction* leads to gradual, incremental changes in thought and behaviour (adapted from the cultural and societal norms) it is through interaction with others that students can examine, clarify and change their conceptual understanding. This theory reminds educators that through the articulation of ideas to others, learners not only receive personalised feedback but are better able to adapt and integrate others' thinking with their own.

Ms Parker's uses peer and teacher *interaction* to develop mathematical understanding. Allowing students to work in groups develops a learning environment reliant on peer *interaction* helping students clarify and construct knowledge. The benefits of cooperative learning range from an increased understanding of the content, improved problem-solving abilities and increased comprehension, to better recall and transfer of complex ideas (Vermette, 1998). The discussion between Terrell and Maurice shows this benefit in action as both used mathematical language to address, and describe, each other's misconceptions. Working on the *social* level, Ms Parker enabled students to develop meaning, ask questions, test hypotheses, and apply the concepts. Since a main principle of Constructivism is that *social interaction* is fundamental for the development of cognition, only through working in small group settings can learning experiences lead to mental development.

Increasingly, teachers will need to make decisions based on solid theoretical frameworks of how people learn. Traditional lecture and rote memorization are no longer adequate pedagogy; as learners of the 21st century must be adequate problem solvers, critical thinkers, and responsible decision makers. Constructivist theory has much to offer.

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## A clue

If you are still struggling with the crossword in **MT217**, Richard Goodman has provided a clue.  
 4 down and 7 across both end with the same number  
 so 15 across and 10 down both end with the same number  
 so 10 across ends with a repeated digit – which means it can only be one of two possible numbers  
**Note: the solution will appear on the website at a later date**

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